

Risk Analysis:

Exercise 1:

- Draw Decision Tree and Calculate EMV
- Which project has lowest risk?

	Investment cost	Good case probability	Good case revenue	Bad case probability	Bad case revenue
Project 1	100000	30%	500000	70%	50000
Project 2	120000	20%	600000	80%	30000
Project 3	150000	25%	1000000	75%	20000

Solution

$$EMV = (P_{\text{good}} \times V_{\text{good}} + P_{\text{bad}} \times V_{\text{bad}}) - \text{Investment}$$

Project 1:

$$EMV = (0.3 \times 500,000 + 0.7 \times 50,000) - 100,000 = 85,000$$

$$\Rightarrow EMV_1 = 85,000$$

Project 2:

$$EMV = (0.2 \times 600,000 + 0.8 \times 30,000) - 120,000 = 24,000$$

$$\Rightarrow EMV_2 = 24,000$$

Project 3:

$$EMV = (0.25 \times 1,000,000 + 0.75 \times 20,000) - 150,000 = 115,000$$

Project 3 has the highest EMV => lowest risk

Project 2 has the lowest EMV => highest risk.